Predicting FIFA World Cup 2026 Finalists Using Machine Learning : Final report

Name: Sindhu G V

Reg no: 24UG00331

Course: CS and AI

Phase 1: Data Preparation and Feature Engineering (Week 1)

The project began with **data collection** from a Kaggle dataset containing team player, performance, and match outcome statistics.

- **Data Cleaning:** No exact duplicate rows were found. Only two columns had missing values (18 missing values each) in the resulting 964 rows.
- **Feature Engineering:** The following five key features were engineered for the classification model:

Feature Name	Description
Goal_Difference	The goal margin for the home team (Home Team Score - Away Team Score).
Home_Team_Avg_Age	The average age of players associated with the home team's country (proxy for experience).
Away_Team_Avg_Age	The average age of players associated with the away team's country (proxy for experience).

Feature Name	Description
Home_Team_Win_Rate	The historical win rate for the home team across all matches in the dataset.
Away_Team_Win_Rate	The historical win rate for the away team across all matches in the dataset.

Phase 2: Model Building and Training (Week 2)

Two classification models were implemented: **Logistic Regression** and **Random Forest**.

- **Initial Performance:** Both models achieved very high initial accuracy, largely because the **Goal_Difference** feature is highly correlated with the win outcome.
- **Preprocessing:** The workflow included data scaling, feature selection, and hyperparameter tuning using 5-fold cross-validation.
- **Hyperparameter Tuning (Random Forest):** The optimal settings for the Random Forest model were determined by the GridSearchCV process.
- **Performance:** The Random Forest Classifier achieved an accuracy of on the test data.

• **Hyperparameter**	• **Best Value**	• **Description**
• classifier__n_estimators	• 50	• The number of decision trees in the forest.

Hyperparameter	Best Value	Description
classifier__max_depth	None	Trees are expanded until all leaves are pure or until all leaves contain less than min_samples_split samples.
classifier__min_samples_split	2	The minimum number of samples required to split an internal node.
feature_selection__threshold	'mean'	Features whose importance score is greater than the **mean** importance of all features were kept.

Phase 3: Model Evaluation and Comparison (Week 3)

The two tuned models were evaluated and critically compared using metrics, Confusion Matrices, and ROC Curves.

1. Model Performance Comparison

Model	Accuracy	ROC-AUC
Random Forest (Tuned)	**0.9862**	**0.9996**
Logistic Regression (Scaled)	**0.9862**	**0.998**

2. Model Trade-offs

Feature	Random Forest (Tuned)	Logistic Regression (Scaled)
ROC-AUC	0.9996 (Superior)	0.9984 (Excellent, but slightly lower)
Interpretability	Low (Black Box)	High (Coefficient show feature direction/impact)
Complexity	High (Ensemble of trees)	Low (Simple linear model)
Training Time	Longer (Training multiple trees)	Shorter (One linear optimization)

- **Conclusion:** The Random Forest has a marginally better **ROC-AUC ()** , making it technically superior for pure prediction accuracy. However, the **Logistic Regression** is the better choice for practical deployment where speed and interpretability matter.

3. Feature Importance and Interpretation

Random Forest (Feature Importance)

Feature	Importance
Home_Team_Win_Rate	0.449603
Goal_Difference	0.287232
Away_Team_Win_Rate	0.160172

Feature	Importance
Home_Team_Avg_Age	0.052670
Away_Team_Avg_Age	0.050323

Logistic Regression (Coefficients)

Feature	Coefficient
Home_Team_Win_Rate	2.610660
Goal_Difference	1.838568
Away_Team_Win_Rate	-1.721453
Home_Team_Avg_Age	0.155845
Away_Team_Avg_Age	-0.155554

- **Interpretation:** Both models overwhelmingly rank **Home_Team_Win_Rate** and **Goal_Difference** as the most influential predictors. The high positive coefficient for **Home_Team_Win_Rate** in Logistic Regression indicates a strong positive correlation with a Home Win. The negative coefficient for **Away_Team_Win_Rate** suggests that a high Away Team Win Rate decreases the probability of a Home Win.

Phase 4: Final Prediction and Reflection (Week 4)

The **Random Forest Classifier** (best ROC-AUC model) was used to predict a simulated 2026 World Cup final between hypothetical finalists, **France and Brazil**.

Simulated Final Prediction

Scenario	**P(Home Win)**
France vs. Brazil (France as "Home")	**0.46**
Brazil vs. France (Brazil as "Home")	**0.82**

- **Predicted Winner: Brazil**. The prediction is based on the team's feature profile, which resulted in a much higher predicted win probability when they were designated as the "Home" team.

Model Limitations and Ethical Considerations

- **Reliance on Historical Data:** The model is overwhelmingly dominated by historical Win Rate and is less sensitive to current team form, injuries, or tactical innovations.
- **Venue Bias:** The model is trained on data with a strong historical home-team bias, which is a structural limitation for neutral-venue World Cup finals.
- **Black Box Nature:** The complexity of the Random Forest makes it difficult to explain *exactly* why Brazil's feature combination led to a probability.
- **Ethical Reporting:** Models should be presented as probabilities, not guarantees, to avoid creating an "Illusion of Certainty" and overshadowing the natural uncertainty and drama of sports.

- **Bias Amplification:** The model may amplify historical biases by favoring traditionally dominant nations, making it harder to predict "underdog" stories.

Application Development

A full application was developed using JavaScript with a robust library for data extraction and Chart.js for visualization. This application simulates the Random Forest training process and executes the final prediction scenario.